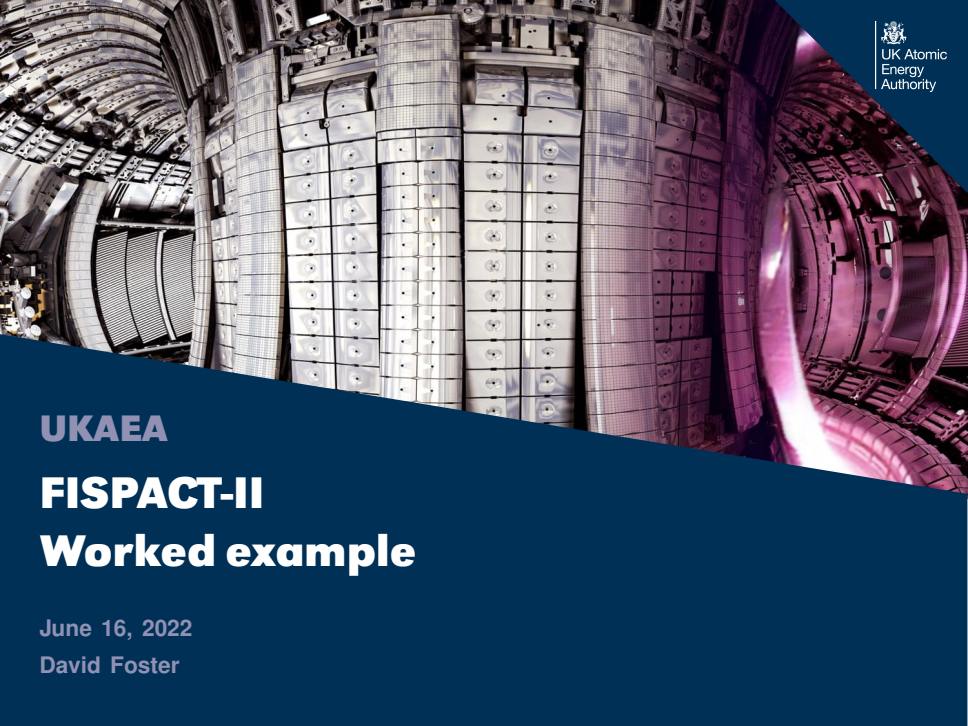




UKAEA

FISPACT-II

Worked example

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Before we can use FISPACT-II we need to install it and unpack some nuclear data.

1. Download and untar Release.tar.bz2:
 - `tar -xvf Release.tar.bz2`
2. Download some nuclear data into the nuclear_data folder
 - <https://fispact.ukaea.uk/nuclear-data/downloads/>
 - `tar -xvf TENDL2017data.tar.bz2`
 - `tar -xvf decay.tar.bz2`
 - `tar -xvf GEFY61data.tar.bz2`
3. Choose the correct FISPACT-II executable

FISPACT-II is a nuclear inventory simulation code .

Worked example: Irradiation of Cobalt for 2 days

1. Configure FISPACT-II to irradiate 1kg of Cobalt and produce a plot of its activity and heat output
2. Use Tendl17, GEFY6.1 and the fluxes from the FNS getting started example
3. Configure one input file (to show a different approach) and a Python3 API approach

Step by step example:

Prerequisite: *Previously ran the FNS example in getting started.*

Step zero

All at the same place

1. Make a new folder called Cobalt_Workshop

```
$mkdir Cobalt_Workshop
```

2. Move into Cobalt_Workshop

```
$cd Cobalt_Workshop
```

Step one

Configure the **files** file: *The files file tells FISPACT-II the location of required inputs for a simulation (page 30 of the 2018 users manual)*

- We need to point FISPACT-II to the correct Tendl-17 files
- ind_nuc, xs_endf need to point at TENDL2017data within nuclear data
- we also need to point the decay data, dk_endf, to decay
- Point fluxes at fluxes

Try doing this. Hint: look at the getting_started examples

ind_nuc	index of nuclides to be included
xs_endf	Cross-section data
fy_endf	Fission yield data
fluxes	Projectile spectrum for a standard group structure
dk_endf	Decay and uncertainty data
collapxi/ collapxo	In/out collapsed cross-section library
absorp	Gamma attenuation library
arrayx	Input and output condensed decay library

```
# files file
```

```
#index of nuclides
```

```
ind_nuc /Users/fosterd/nuclear_data/TENDL2017
```

```
#Lib of cross sections
```

```
xs_endf /Users/fosterd/nuclear_data/TENDL2017data/tal
```

```
# Library fission data
```

```
fy_endf /Users/fosterd/nuclear_data/GEFY61data/gefy61
```

```
#fluxes
```

```
fluxes fluxes
```

```
# lib decay data
```

```
dk_endf /Users/fosterd/nuclear_data/decay/decay_2012
```

```
# collapsed cross sections data
```

```
collapxi COLLAPX
```

```
collapxo COLLAPX
```

```
# gamma attenuation data
```

```
absorp /Users/fosterd/nuclear_data/decay/abs_2012
```

```
# condensed decay and fission data (in and out)
```

```
arrayx ARRAYX
```

```
# files file

#index of nuclides
ind_nuc /Users/fosterd/nuclear_data/TENDL2017data/tendl17_decay12_index

#Lib of cross sections
xs_endf /Users/fosterd/nuclear_data/TENDL2017data/tal2017-n/gxs-709
# Library fission data
fy_endf /Users/fosterd/nuclear_data/GEFY61data/gefy61_nfy

#fluxes
fluxes fluxes

# lib decay data
dk_endf /Users/fosterd/nuclear_data/decay/decay_2012

# collapsed cross sections data
collapxi COLLAPX
collapxo COLLAPX

# gamma attenuation data
absorp /Users/fosterd/nuclear_data/decay/abs_2012

# condensed decay and fission data (in and out)
arrayx ARRAYX
```

Step two: input file

Your files file should look like mine

The input file tells FISPACT-II what to do.

In the getting started examples we see that the input files are separated into four separate *files*, *collapse*, *condense*, *print_lib*, *inventory*.

Here we are making one file to do it all, called **combine.i**

Step two: input file

Required components of the input file.

Initial settings	
CLOBBER	Allows Fispact-II to overwrite output files
GETXS 1 709	(1) tells Fispact to collapse the cross sections from the ENDF library, (709) states that the 709 group energy multi-group should be used
GETDECAY 1	causes the decay data to be read from the decay library files connected to dk endf
FISPACT	separates the library input from the initial conditions
* Cobalt	key word information about the particular run - used in graph titles

Step two: input file

Material definition	
MASS 1.0 1	total mass (kg) and the number of elements in the material to be irradiated
CO 100.0	100% Cobalt

Output settings	
GRAPH 2 2 1 1 3	Graph output settings. (2) plots, (2) use gnu-plot, (1) include uncertainty data, (1) activity, (3) Heat output
HALF	include half-life of each nuclide in output

Step two: input file

Irradiation phase	
FLUX 10.0E+10	total energy-integrated projectile flux (in $\text{cm}^{-2}\text{s}^{-1}$)
ATOMS	starts the initial solution of the inventory equations
TIME 2 DAYS	sets the first time interval for the inventory calculation and terminates the initial conditions section
ATOMS	restarts the solution of the inventory equations

Step two: input file

cooling phase	
FLUX 0.	total energy-integrated projectile flux (in $\text{cm}^{-2}\text{s}^{-1}$) for this time period
ZERO	reset the time value to zero after an irradiation
TIME 1 DAYS ATOMS	Inventories after 1 day are output
END	terminates the input of data for a particular run. Must be preceded by the * with a title.



```
CLOBBER
MONITOR 1
GETXS -1 709
GETDECAY 1
FISPACT
* Cobalt
TOLERANCE 0 5.0E2 2.0E-6
MASS 1.0 1
CO 100.0
GRAPH 2 2 1 1 3
HALF
<< -----irradiation phase----- >>
FLUX 10.0E+10
ATOMS
TIME 2 DAYS
ATOMS
<< -----cooling phase----- >>
FLUX 0.
ZERO
TIME 1 HOURS ATOMS
TIME 2 HOURS ATOMS
TIME 3 HOURS ATOMS
TIME 4 HOURS ATOMS
TIME 5 HOURS ATOMS
TIME 1 DAYS ATOMS
TIME 1 DAYS ATOMS
TIME 1 DAYS ATOMS
TIME 1 DAYS ATOMS
END
* END
```

Step three: Run FISPACT-II

Now we have the input files we simply type:

```
$/ Pathto / fispact combine
```

after a few mins this will produce a print out like:

```
combine:  cpu time =  530.      secs.  1196 errors/warnings , for details see log file .
```

and a file called **combine.gra**.

Step three: Output, combine.gra

```
# Run timestamp = 12:13:52 22 October 2021
# Cobalt
# Time after irradiation (years)
#           Activity (Bq/k)
# index      0
#   time      value      uncert
0.00000E+00  4.63955E+11  0.00000E+00
1.14077E-04  4.15897E+11  0.00000E+00
3.42231E-04  3.51971E+11  0.00000E+00
6.84463E-04  2.77312E+11  0.00000E+00
1.14077E-03  2.04769E+11  0.00000E+00
1.71116E-03  1.42601E+11  0.00000E+00
4.44901E-03  3.41037E+10  0.00000E+00
7.18686E-03  1.73200E+10  0.00000E+00
9.92471E-03  1.46079E+10  0.00000E+00
1.26626E-02  1.40675E+10  0.00000E+00
```

```
# index      1
# half-life  value      name
1.2182E-01  1.3022E+09  ^{59}Fe
1.9400E-01  1.1193E+10  ^{58}Co
2.9459E-04  2.8294E+10  ^{56}Mn
1.0153E-03  4.1229E+11  ^{58m}Co
```

Step three: Plotting

Now we have the **.gra** file we can produce a plot.

gnuplot

On Linux and Mac (if installed) we can simply type:

```
$gnuplot combine.plt
```

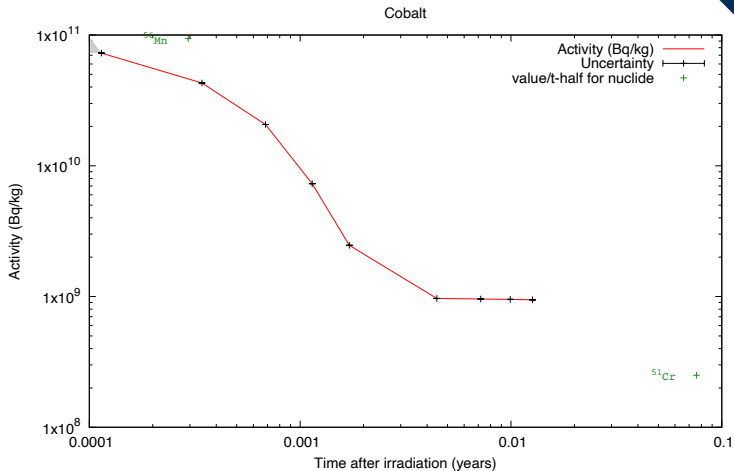
Producing, a file called **combine.gra.ps**

Matplotlib

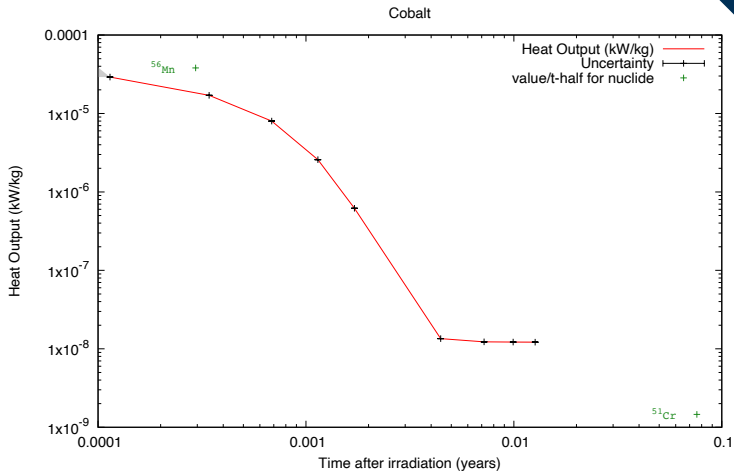
Alternatively if we change *GRAPH 2 2 1 1 3* to *GRAPH 2 3 1 1 3* a Python3 Matplotlib file will be produced:

```
$python3 combine.py
```

Producing, a file called **combine.pdf**

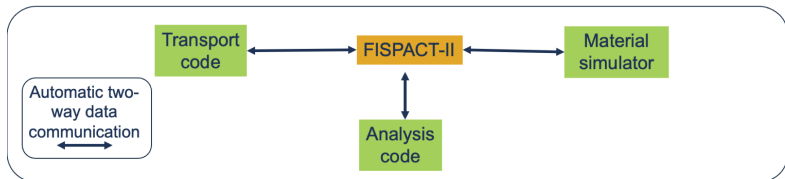


file name = combine.gra run timestamp = 11:53:55 12 October 2021



file name = combine.gra run timestamp = 11:53:55 12 October 2021

- FISPACT-II v 5 now has an API (Application Programming Interface)
- Can now 'plug' into codes (Python, Fortran, C, C++)
- Help optimise workflow
- Shown to give a 10x speed up when using TENDL (one example)



FISPACT-II API python3 example

Nuclear data

- Define FISPACT nuclear data object
- Define the location of the nuclear data
- Load the nuclear data

Input

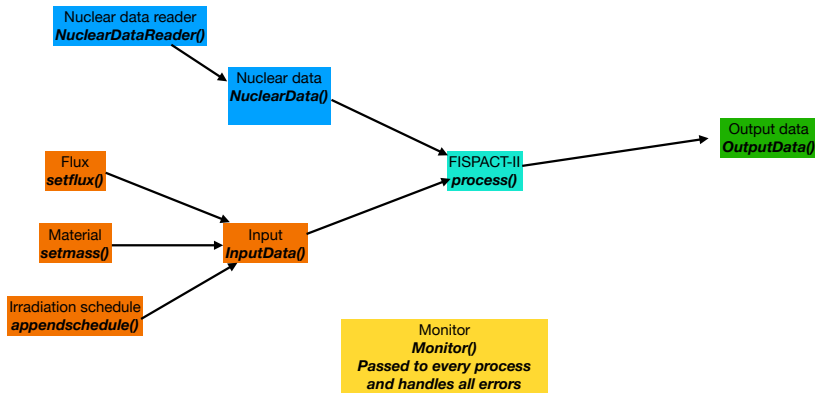
- Define the FISPACT input data object
- Set the flux and the material definition
- Set the radiation/cooling schedule

run

- Pass the nuclear data and input objects to FISPACT-II
 - `pf.process(i, nd, o, m, op=computefunc)`

Post process

FISPACT API data flow



- We now have the Activation and Heat output

Extensions

- Change the irradiation time, and then adding those new curves to the plots. How does the activity change with increased irradiation time?
- Change the neutron flux.
- Did we need to include the GEFY fission data?